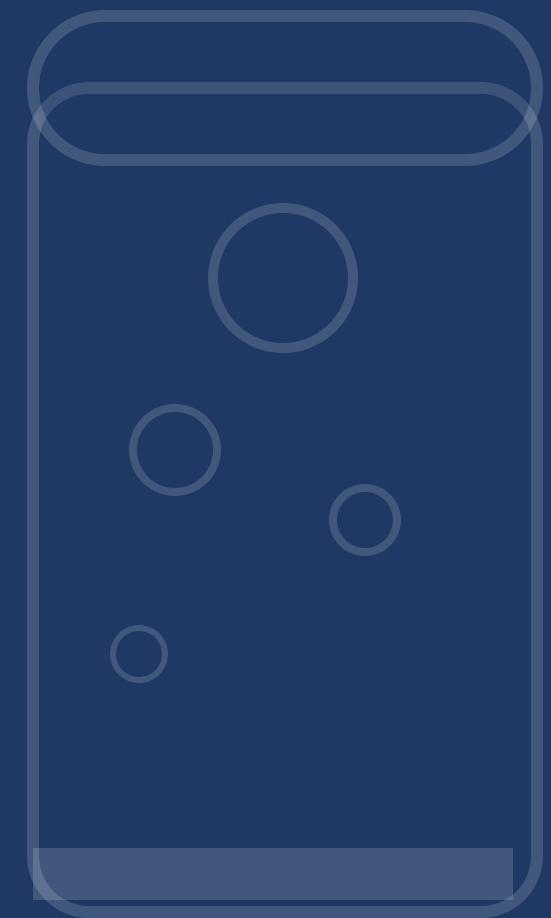


COURSE PROJECT KICKOFF

CL 420: Biochemical Engineering

Course Project — Design of an Upstream Bioprocess

Design a real bioprocess. Not a literature review.



Teams and Product Choice

Teams

Fixed teams of 3.

No exceptions.

Pick ONE Product

MODERN BIOPHARMACEUTICALS

mAb

Recombinant Insulin

Recombinant Enzyme

Vaccine Antigen Antigen

AAV

GLP-1 Peptide

CLASSIC FERMENTATION PRODUCTS

Penicillin

Citric Acid

Lactic Acid

Lysine

Ethanol

No two teams may share a product/host combination — register first-come, first-served. Something not on this list? Get it approved first.

What Your Report Must Cover

1

Product & Process Definition

What, why, host organism, CQAs

2

Biological System

Growth/product formation, pathways, substrates

3

Bioreactor & Upstream Process

Configuration, operating conditions, feeding strategy

4

Engineering Calculations

Material balances, kinetics, kLa, reactor sizing

5

Process Optimization

Bottlenecks, key variables, improvement strategy

6

Downstream Processing

High level only

7

Preliminary Economics

Order-of-magnitude cost drivers

8

Final Recommendation

Process choice, assumptions, risks

Think Like a Biochemical Engineer

TITER

Product per litre

g / L

RATE

How fast you make it

g / L / h

YIELD

Substrate → product efficiency
efficiency

CQA — Critical Quality Attribute

A product property that must stay in spec — e.g. glycosylation, potency, purity.

PAT — Process Analytical Technology

Monitoring and controlling the process in real time to keep CQAs on target.

Every calculation in your report should feed back into these numbers.

COURSE PROJECT KICKOFF

Engineering Calculations — What We Actually Need to See

- ✓ Material balance over the bioreactor (substrate, biomass, product, by-products)
- ✓ Growth & product-formation kinetics (Monod, Luedeking–Piret, or equivalent) with parameters
- ✓ Yield coefficients ($Y_{x/s}$, $Y_{p/s}$)
- ✓ Oxygen transfer: estimated k_La , oxygen demand (OUR), agitation/aeration required
- ✓ agitation/aeration required
- ✓ Heat generation/removal estimate (where relevant)
- ✓ Reactor volume estimate
- ✓ Final predicted TITER, RATE, and YIELD stated explicitly

Show your working — equations, parameter values with source, and the resulting numbers. Not just a final answer.

Format and Submission



One PDF via the course portal, by the announced deadline (standard late policy applies)



Cover page with title, team names, and host organism



Cite all sources used for parameters or process data



Everyone on the team must speak at both presentations — individual contribution and Q&A may factor into the grade into the grade



10-page ceiling for the final report — a tight, calculation-dense 7 pages beats a padded 10

How the Project Is Evaluated



What Earns Marks (and What Doesn't)

Earns marks

- ✓ Real material balances, kinetics, $k_L a$ / oxygen-demand calculations with working shown
- ✓ Clearly stated, justified assumptions
- ✓ A specific, quantified optimization idea
- ✓ A clear final recommendation with risks and assumptions spelled out

Won't earn marks

- ✗ Retelling textbook / review content with no numbers of your own
- ✗ Numbers presented with no basis
- ✗ Generic suggestions like "optimize the conditions"
- ✗ A report that never commits to a process